A surrogate model for the transient temperature field around an emplaced waste canister

THERESA 1.0.0 — theory*

Abstract

A three-dimensional transient heat-conduction model of a Pollux-sized canister in drift emplacement takes minutes per design. This note derives the construction that reduces a parameter study to a single interpolation. Because the governing equation is linear, the surface temperature, the geothermal gradient, the emplacement depth, the canister loading and the decay-heat curve — its amplitudes, its decay constants and their number — enter *in closed form*. The ten parameters that change the differential operator — the two spacings and the conductivity and volumetric heat capacity of four materials — are interpolated. The construction agrees with a conventional simulation to 0.07 K at the peak, the interpolation with unseen designs to 0.130 K.

1 The model

1.1 Domain

The repository is idealised as an infinite array of identical, simultaneously emplaced canisters lying in parallel drifts. The array has four planes of mirror symmetry through each canister, so it suffices to model one quarter of the unit cell,

$$\Omega = \left[0, \tfrac{p}{2}\right] \times \left[0, \tfrac{a}{2}\right] \times \left[-H,\, 0\right],$$

where x runs along the drift axis, y perpendicular to it, and z is height with $z = 0$ at ground level. The two lateral extents are half the canister pitch p and half the drift spacing a.

Neighbouring canisters are separated by a *clear canister spacing* g, measured between their end faces, from which the pitch follows:

$$p = L + g, \tag{1}$$

with L the canister length. This matters for more than nomenclature: a pitch would have to satisfy $p > L$, so treating p and L as independent design variables would make the design space non-rectangular and generate impossible designs in one corner. In (g, L) the box is rectangular by construction.

Ω contains four subdomains with distinct, constant material properties: the canister (a quarter cylinder of radius r_c and length $L/2$ within Ω), the backfill, the liner, and the host rock. The drift runs the full length of the cell; only the canister is finite.

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1.2 Governing problem

With ρ, c_p and λ piecewise constant,

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) + q, \qquad \text{in } \Omega \times (0, t_{\text{end}}], \tag{2}$$

subject to

$$\begin{aligned} T &= T_{\text{top}} && \text{on } \Gamma_{\text{top}} \ (z = 0), \\ T &= T_{\text{top}} + \gamma H && \text{on } \Gamma_{\text{bot}} \ (z = -H), \\ \lambda \nabla T \cdot n &= 0 && \text{on } \Gamma_{\text{sym}}, \end{aligned} \tag{3}$$

where γ is the geothermal gradient and Γ_{sym} collects the four lateral faces. A zero-flux condition is the natural boundary condition of the weak form, so in a finite-element discretisation the symmetry planes are imposed by *not* constraining them.

The source is the canister's decay heat, distributed uniformly over the canister volume:

$$q(x,t) = \begin{cases} \dfrac{Q(t)}{4\,V_{\text{c}}}, & x \in \Omega_{\text{c}}, \\ 0, & \text{otherwise}, \end{cases} \qquad Q(t) = \sum_{i=1}^{N} P_i \, e^{-\lambda_i t}. \tag{4}$$

Here Q is the power of the *whole* canister and V_{c} the volume of the quarter canister actually present in Ω, so that $\int_\Omega q \, \mathrm{d}V = Q/4$: the quarter cell receives a quarter of the power, as it must, since it represents a quarter of the unit cell. The factor is easy to get wrong by four.

A second subtlety hides in V_{c}. A tetrahedral mesh approximates the cylinder from the inside, so the meshed canister is smaller than the analytic one — about 0.45 % at the resolution used here. Since the solver integrates q over the *meshed* elements, referring q to the analytic volume would inject that much too little heat, worth about a kelvin on a rise of a few hundred. V_{c} is therefore the meshed volume.

1.3 Initial condition

The simulation starts from the undisturbed rock, i.e. from the source-free steady state T_{ss} solving

$$\nabla \cdot (\lambda \nabla T_{\text{ss}}) = 0 \tag{5}$$

with the same boundary conditions (3).

2 Linearity

Equation (2) is linear in T: the coefficients ρ, c_p and λ do not depend on T. Every reduction below follows from that single fact, and every one of them fails the moment a temperature-dependent conductivity is introduced.

2.1 The steady state: surface temperature and gradient are free

Proposition 1. *Let ψ solve (5) with $\psi = 0$ on Γ_{top}, $\psi = H$ on Γ_{bot}, and zero flux on Γ_{sym}. Then for every T_{top} and every γ,*

$$T_{\text{ss}}(x) = T_{\text{top}} + \gamma \, \psi(x). \tag{6}$$

Proof. The right-hand side satisfies (5), because T_{top} is a constant and $\nabla \cdot (\lambda \nabla (T_{\text{top}} + \gamma\psi)) = \gamma \, \nabla \cdot (\lambda \nabla \psi) = 0$. It takes the value T_{top} on Γ_{top} and $T_{\text{top}} + \gamma H$ on Γ_{bot}, and its normal flux vanishes on Γ_{sym}. The elliptic problem with mixed Dirichlet–Neumann data has a unique solution, so the two coincide. □

ψ depends on the geometry alone. One source-free solve per design therefore covers *every* surface temperature and *every* geothermal gradient exactly, at no further cost. In a homogeneous medium $\psi(x) = -z$; the material contrast perturbs it only locally, by about 0.02 K-equivalent at the canister, and computing ψ rather than assuming a linear profile keeps that correction.

2.2 The heating: the decay curve is free

Write $u = T - T_{\mathrm{ss}}$. Subtracting (5) from (2) and using that T_{ss} is time-independent,

$$\rho c_p \frac{\partial u}{\partial t} = \nabla \cdot (\lambda \nabla u) + q, \qquad u(\cdot, 0) = 0, \tag{7}$$

with $u = 0$ on $\Gamma_{\mathrm{top}} \cup \Gamma_{\mathrm{bot}}$ and zero flux on Γ_{sym}. The boundary data of u are homogeneous, so u *does not depend on* T_{top} *or* γ *at all.* Together with (6),

$$T(x,t) = \underbrace{T_{\mathrm{top}} + \gamma\,\psi(x)}_{\text{one steady solve}} + \underbrace{u(x,t)}_{\text{one transient solve}} . \tag{8}$$

Problem (7) is linear and time-invariant in the source. Let $S(x,t)$ denote its solution for the unit step $Q \equiv 1\,\mathrm{W}$. Then Duhamel's principle gives, for any power history Q,

$$u(x,t) = \int_0^t \dot{S}(x, t-\tau)\, Q(\tau)\, \mathrm{d}\tau = \int_0^t \dot{S}(x,v)\, Q(t-v)\, \mathrm{d}v. \tag{9}$$

Evaluating (9) exactly

S is known at the solver's own time levels $0 = t_0 < t_1 < \cdots < t_{N_t}$. Between them the finite-element solution is interpolated linearly in time, so taking S piecewise linear is not an approximation *of the data* — it is the data. Its derivative is then piecewise constant,

$$\dot{S}(v) = s_j \equiv \frac{S_{j+1} - S_j}{t_{j+1} - t_j}, \qquad v \in [t_j, t_{j+1}),$$

and each interval integrates in closed form. For a single source term $Pe^{-\lambda t}$ and an output time t_n,

$$\int_{t_j}^{t_{j+1}} P\, e^{-\lambda(t_n - v)}\, \mathrm{d}v = \frac{P}{\lambda}\Big(e^{-\lambda(t_n - t_{j+1})} - e^{-\lambda(t_n - t_j)}\Big),$$

so that, summing over the terms and the intervals below t_n,

$$\boxed{u(t_n) = \sum_{i=1}^{N} \frac{P_i}{\lambda_i} \sum_{j<n} s_j \Big(e^{-\lambda_i(t_n - t_{j+1})} - e^{-\lambda_i(t_n - t_j)}\Big).} \tag{10}$$

Every exponent in (10) is non-positive. The algebraically equivalent form with $e^{+\lambda_i v}$ would overflow in double precision for a short-lived nuclide over a 1000 a window; this one cannot.

Two limiting cases are worth recording. For $\lambda_i \to 0$ the quotient tends to the interval length $t_{j+1} - t_j$, and (10) collapses to $u(t_n) = P\,S(t_n)$ — a constant source reproduces the step response, which is the identity that fixes the normalisation of S and is checked to round-off in the test suite. And if S itself happens to be an exponential sum, $S(t) = \sum_k a_k(1 - e^{-\mu_k t})$, term-by-term integration of (9) gives the independent closed form

$$u(t) = \sum_k \sum_i a_k \mu_k P_i\, \frac{e^{-\lambda_i t} - e^{-\mu_k t}}{\mu_k - \lambda_i}, \tag{11}$$

with the removable singularity at $\mu_k = \lambda_i$ replaced by its limit $a_k \mu_k P_i\, t\, e^{-\mu_k t}$. Equation (11) is the reference against which (10) is verified on a kernel whose exact answer is known.

Consequence

The amplitudes P_i, the decay constants λ_i and the term count N appear only in (10), never in a simulation. Six exponential terms cost exactly what four or twenty cost, and introduce no error of their own. The interim storage time t_s is likewise free: shifting an exponential in time only rescales it,

$$Q(t) = n \sum_i p_i e^{-\lambda_i (t + t_\mathrm{s})} = \sum_i \underbrace{n\, p_i\, e^{-\lambda_i t_\mathrm{s}}}_{P_i} e^{-\lambda_i t}, \tag{12}$$

so it folds entirely into the amplitudes.

2.3 The loading, forwards and backwards

From (7), scaling the source scales the response: $Q \mapsto \alpha Q$ implies $u \mapsto \alpha u$. The loading — as a package count n in (12), or directly as a power — is therefore a multiplication.

The inverse question is the one a design review asks, and it is equally exact. Let Γ_c be the canister surface and T_lim a design limit. Requiring $T \le T_\mathrm{lim}$ everywhere on Γ_c and at all times gives, using (8),

$$T_\mathrm{ss}(x) + \alpha\, u(x,t) \le T_\mathrm{lim} \qquad \forall\, x \in \Gamma_\mathrm{c},\ t \in [0, t_\mathrm{end}],$$

and since $u \ge 0$,

$$\boxed{\alpha_\mathrm{max} = \min_{\substack{x \in \Gamma_\mathrm{c},\ t \\ u(x,t) > 0}} \frac{T_\mathrm{lim} - T_\mathrm{ss}(x)}{u(x,t)}.} \tag{13}$$

The largest admissible loading is $\alpha_\mathrm{max} Q(0)$. No search, no iteration, no further simulation: one pass over the stored (x,t) grid. If $T_\mathrm{ss} > T_\mathrm{lim}$ anywhere on the canister surface — the undisturbed rock is already too hot — the constraint set is empty and no loading works, which (13) reports as $\alpha_\mathrm{max} \le 0$.

Note that the maximum over the surface is taken *after* the convolution, per time step, rather than maximising the step response first and convolving a single curve.

The leading point on the surface does change with time — for the reference design it is the barrel at the mid-length plane for 292 of 296 time steps, and one neighbouring point for the other four. What makes this harmless is that the two leading points are never more than 1.3×10^{-3} apart in relative terms, so the two orders were measured to agree to better than 0.1 mK. Convolving first is therefore not a correction to a present error but insurance: it costs nothing here, and it remains correct if the range is ever widened to shapes where the leading points separate, where maximising first would overestimate.

2.4 Depth

The emplacement depth d acts twice. It sets the undisturbed temperature at the canister, which is $T_\mathrm{top} + \gamma d$ to within the local material perturbation and is carried exactly by ψ in (6). And it sets the distance from the heated zone to the upper Dirichlet boundary, which can only affect u.

The second effect is bounded by diffusion. With $\alpha_\mathrm{th} = \lambda / (\rho c_p) \approx 8.6 \times 10^{-7}\,\mathrm{m^2/s}$ for the host rock, the thermal penetration depth after 1000 a is $\sqrt{\alpha_\mathrm{th} t} \approx 180$ m, and at the time of the peak only 35 – 55 m. At 300 m of cover or more the boundary is several penetration depths away.

This is an estimate, not a proof, so the model measures it: kernels computed at $d \in \{300, 500, 700, 1000\}$ m with everything else held fixed differ, after the analytic term is removed,

by at most 0.062 K (Table 1). Depth is therefore treated analytically and is not interpolated. The vertical extent keeps a constant clearance below the drift, so that the comparison isolates the upper boundary.

3 Interpolation of the step response

After Sections 2.1–2.4, the only quantities that still require simulation are those that change the differential operator itself:

$$\pi = (g,\ a,\ \{\lambda_m\}_{m=1}^{4},\ \{(\rho c)_m\}_{m=1}^{4}) \in \mathbb{R}^{10},$$

the clear canister spacing and the drift spacing, which change the mesh, and the thermal conductivity and volumetric heat capacity of the four materials, which change the coefficients. The object to be learned is the map $\pi \mapsto S(\cdot, \cdot; \pi)$.

Two remarks on this split. First, heat capacity appears only as the product ρc: the density and the specific heat never enter (2) separately, so treating them as two parameters would add a dimension carrying no information. Second, the split is not between "important" and "unimportant" but between what the designer *chooses* and what the operator *contains*. Everything exact in Sections 2.1–2.4 is a choice: a surface temperature, a loading, a storage time. Everything interpolated is a property. That the second group is also where the physical uncertainty lives is what makes a variance-based sensitivity analysis over π meaningful.

The canister length L is deliberately *not* in π. A canister type is a discrete choice, and a length interpolated between two types corresponds to no object; a different type is trained separately.

3.1 Sampling the canister surface

S is needed on the canister surface, but the finite-element node sets of two different designs are not comparable, so a nodal representation cannot be interpolated across designs. Instead S is sampled at $N_s = 156$ probe points fixed in *canister-local* coordinates: a grid in the circumferential angle $\theta \in [0, \pi]$ and the normalised axial coordinate $\xi \in [0, 1]$ of the half-length, plus rings on the end face. Because the grid is normalised, a probe sits at the same relative position for every design, so the N_s components of S mean the same thing from one design to the next — which is what makes interpolation across designs well defined at all.

The sampling loss of that grid is measured against the exact nodal maximum over the canister-wall submesh, and is recorded for every training run. It is $-0.01\,\%$: the probe grid is marginally *above* the nodal maximum, because the probes sit 2 cm inside the surface, where the cast iron is a shade hotter.

3.2 Reduced basis

A kernel is an $N_t \times N_s$ array — some 4.6×10^4 numbers — so fitting one regressor per entry is neither necessary nor sensible. Kernels of different designs are very similar: they share a large near-field part, the conduction drop through the backfill and liner immediately around the canister, which depends on the spacings not at all and on L only through $1/L$; and they differ in a smaller, smooth array-interaction part.

A singular value decomposition of the mean-centred training kernels separates exactly that. Writing $K(\pi) \in \mathbb{R}^{N_t \times N_s}$ flattened,

$$K(\pi) \ \approx\ \bar{K} + \sum_{k=1}^{M} c_k(\pi)\, \Phi_k, \tag{14}$$

with Φ_k the leading right singular vectors. Modes are retained up to a captured energy fraction of $1 - 10^{-8}$, at most $M = 8$; the remaining modes carry run-to-run scatter only.

3.3 Regression

Each coefficient c_k is a smooth scalar function of the ten variables, fitted by a Gaussian process with a Matérn-5/2 covariance with one length scale per input (ARD) and a white-noise term.

The inputs are transformed before fitting, conductivities and spacings reciprocally and volumetric heat capacities directly:

$$z = \big(1/g,\ 1/a,\ \{1/\lambda_m\}_{m=1}^{4},\ \{(\rho c)_m\}_{m=1}^{4}\big). \tag{15}$$

For the two spacings this reflects that neighbour interaction scales with inverse distance. For the conductivities it reflects that a conduction resistance is proportional to $1/\lambda$, so the response is close to linear in $1/\lambda$ and far from it in λ. A volumetric heat capacity is not a resistance — it sets how fast the near field charges up — and enters roughly linearly, so it is sampled as it stands.

Training points are drawn from a Latin hypercube rather than a tensor grid. At ten dimensions that is a necessity rather than a preference: six levels per variable would be $6^{10} \approx 6 \times 10^7$ runs, whereas n scattered points give n distinct levels in *every* variable. Sampling is done in the transformed coordinates (15).

3.4 Uncertainty

Each Gaussian process returns a posterior standard deviation σ_k for its coefficient. Every step from there to a temperature is linear — reconstruction (14), convolution (10), and the additive split (8) — so a one-sigma perturbation of c_k maps to a temperature curve $\delta T_k(t)$ by the same operations. The coefficients are fitted independently, so the contributions combine in quadrature:

$$\sigma_T(t) = \Big(\sum_{k=1}^{M} \delta T_k(t)^2\Big)^{1/2}, \qquad \delta T_k = \mathcal{D}\big[\sigma_k \Phi_k\big], \tag{16}$$

where $\mathcal{D}$ denotes the convolution operator of (10) evaluated at the probe point that is hottest at time t.

This is the uncertainty of the *interpolation*, and of nothing else. It says how unsure the regression is about a design it was not trained on. It does not include the uncertainty of the physical model, which is dominated by the material data and the modelling assumptions of Section 5 and is very much larger.

4 Discretisation and error budget

4.1 Time

The schedule is blocked-logarithmic: a few steps at one size, then the size grown by a fixed factor, until a ceiling is reached. Two things motivate this rather than a smoothly growing schedule. The step response rises over six decades of time and needs resolution at the fast end; and, since the problem is linear, the solver may reuse its setup for as long as the step size is unchanged, so keeping the number of *distinct* sizes small is what makes the run cheap. The shipped schedule has 295 steps but only 28 distinct sizes, so 9 % of steps pay for a solver setup.

4.2 Space and solver

Element size is graded from 0.15 m at the canister to 30 m in the far field, reaching the coarse size only after 200 m. The last figure is not cosmetic: the field to be resolved extends about 180 m. The grading meshes every corner of the design space.

Uniform refinement of this grading by $\sqrt{2}$ over four levels gives an observed order of convergence of 2.22 and a Richardson estimate 0.284 K above the shipped result, the approach being monotone from below. The same exercise in time — refining the whole schedule self-similarly — moves the peak by 0.004 K with the sign alternating, which is the solver's noise floor rather than a convergence rate. The two are not comparable, and the shipped configuration is accordingly generous in time and tight in space.

The linear systems are solved by conjugate gradients with diagonal preconditioning. The system is symmetric positive definite — the textbook case — and a direct sparse factorisation was measured to be at least 3.7× slower at this problem size.

4.3 Budget

Table 1: Measured error contributions and their references. Accuracy target: 0.5 K on the peak temperature.

Contribution	Measured against	Error / K
Linear construction, Eq. (8)–(10)	conventional simulation	0.070
Interpolation of the kernel	unseen simulated designs	0.130
Treating depth analytically	kernels at four depths	0.062
Time discretisation of (10)	analytic kernel, Eq. (11)	0.017
Probe-grid sampling	exact nodal maximum	0.100
Spatial discretisation	Richardson extrapolation	0.284

The first row is the decisive one. It compares the assembled answer — one steady solve, one step-response run, and the convolution with a six-term decay curve — against a conventional transient simulation of the same case with the real decay curve as its source and the undisturbed field as its initial condition. The two agree to 0.070 K at the peak. Everything in Section 2 rests on that agreement.

The last row is the largest, and it is of a different kind. Every other entry is a difference between two things computed on the same mesh, so the mesh cancels out of it; the spatial term is the distance from the mesh to the continuum, and nothing in the other measurements can see it. It is reported here as a *documented offset* rather than applied as a correction. It is systematic and one-signed — all designs use the same mesh and all underestimate by about the same amount — so it largely cancels in comparisons between designs, which is what the model is mostly used for. It does not cancel in an absolute value compared against a temperature limit. Applying the extrapolated offset as a constant would trade a known bias for an unknown one: it was measured at a single design point, and there is no evidence that it is constant over a ten-dimensional parameter space. An absolute peak needed to better than a few tenths of a kelvin is best obtained from one directly simulated run on a refined mesh at that design point.

5 Assumptions and limits

- Pure heat conduction. No convection, no hydraulics, no mechanics, no radiative exchange across gaps, no resaturation of the backfill.

- Temperature-independent material properties. This is what makes the superposition exact, and it is the assumption that would have to go first if the model were extended: a conductivity $\lambda(T)$ invalidates Proposition 1, Duhamel's principle (9) and the loading relation (13) alike.
- Perfect thermal contact between all regions; no contact resistance.
- An infinite emplacement field with all canisters emplaced simultaneously — the consequence of symmetry on all four lateral faces, and the conservative standard assumption for near-field design. A genuinely isolated canister would require far-field Dirichlet boundaries instead.
- The heat source is uniform over the canister volume.
- The host rock is homogeneous over the whole column: no geological layering, no depth-dependent conductivity or gradient.
- The interpolation is valid over the trained ranges of g, a and L; outside them the prediction is an extrapolation and reports itself as one.
- THERESA is a private hobby project. Geometry, material parameters, decay-heat spectra and the design limit are simplified or assumed values, loosely based on published literature; they are not realistic and do not describe any existing or planned repository. The results are not suitable for design, licensing or safety assessments.